// ORDER FLOW RESEARCH SERIES · N-01

DO LARGE TRADES PREDICT PRICE.

A statistical study of the relationship between the largest trade printed in each minute of the CME tape and the direction of NQ futures over the minutes that follow. Nine years of official Level 3 (MBO) data, three million one-minute bars, more than three million simulated trades. Executed end to end in MultiCharts through The Bridge.

INSTRUMENT
NQ Futures

SAMPLE
May 2017 – Jul 2026

DATA
CME MBO / Level 3

BARS
~3.0M x 1-minute

METHOD
Fixed-horizon events

PLATFORM
MultiCharts + The Bridge

PREPARED BY
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ISSUE
v1.0 / 2026

STATUS
Members Release

PROCESS > PREDICTION

// ABSTRACT

THE CASE.

Retail trading culture treats the candle as the atom of information. This study asks whether the tape underneath the candle knows more: specifically, whether the single largest trade of each minute, read directly from the exchange order book, predicts where price goes next.

We measure everything the same way. A signal on bar t opens a position at the open of bar t+1 and closes it exactly N bars later. Every strategy is therefore a stream of identical fixed-length bets, and average profit per trade becomes a direct measure of information content. All headline tests are gross of costs by design: this is a study of predictability first, tradability second.

The answer is layered. Large trades carry roughly five times the information of the candle itself. The information needs about a quarter of an hour to fully express. It lives almost entirely inside cash hours. It concentrates in prints of 70 to 190 contracts, in prints located in the candle's wick, and in prints that arrive when the resting book is balanced. Stacked into one configuration, the result is a signal that clears realistic costs across a wide parameter band.

// HEADLINE FINDINGS

Candle baseline, 1-min hold	$0.41 / trade
Large trade (≥100), 1-min hold	$1.88 / trade
Information multiple	4.6x
Time to full expression	~15–20 min
Share of prints in RTH	79%
Final configuration, avg trade	$26.85
Final configuration, PF	1.13

> *The edge is not in the color of the candle. It is in the size, the location, and the context of the print inside it.*

HEADLINE COMMENTARY

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01

// OFFICIAL EXCHANGE DATA

DATA & METHOD.

// The data

Every number in this paper is computed from **official CME Market-By-Order (MBO) Level 3 data**: the complete order-by-order record of the NQ book, May 21, 2017 to July 6, 2026. This is the same data class institutional desks buy. From it, The Bridge reconstructs the full order book and serves condensed per-minute features into MultiCharts, where every test below is a plain PowerLanguage backtest.

The feature at the center of this study is the **largest single trade of each minute**: its signed size (buyer- or seller-initiated, from the aggressor flag on the tape) and the exact price where it happened. No approximation from volume bars, no tick-rule guessing. The tape itself.

// The measurement

Every test uses the same fixed-horizon design. A condition observed on bar t opens a position at the open of bar t+1 and closes it exactly HoldBars later. Consecutive signals produce separate, back-to-back trades. Every trade is therefore a bet of identical length, so **average profit per trade is a clean estimate of information content**, comparable across strategies.

Headline tests run gross of commission and slippage, with a realistic round-trip friction of about $14 per trade drawn as a reference line wherever it matters. The final configuration is additionally assessed net of that friction in section 12.



FIG. 1 · THE RAW MATERIAL: EVERY MINUTE'S LARGEST PRINT, SIZED AND SIDED, ON THE NQ CHART

// SAME-SOURCE RULE

Chart candles and served features are built from the same Databento MBO record, spliced per contract on the same roll. The Bridge ships an alignment verifier that scores the join bar by bar; the dataset used here verifies at the expected ~95% match level. Features and candles describe the same minute of the same contract, or the test does not run.

// CONVENTIONS

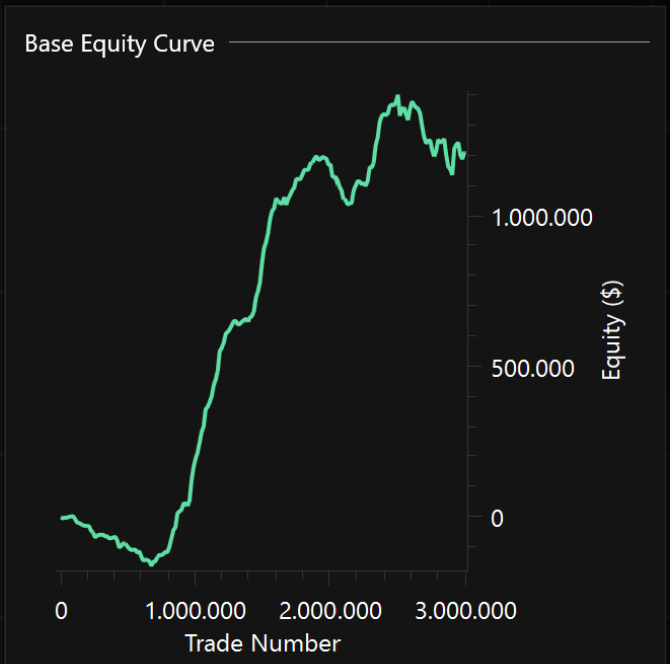
All times Eastern. A "whale" is a print at or above the MinSize threshold, 100 contracts unless stated. Dollar figures are per single NQ contract ($20 per point). Percent profitable is reported but the edge in this study is carried by payoff asymmetry, not hit rate.

02

// REPRODUCIBLE PIPELINE

WHAT WE DID, STEP BY STEP.

1. **Built the dataset.** Nine years of NQ MBO history fetched contract by contract, roll-spliced into one continuous feature file and one matching price series, both from the same tape.
2. **Verified alignment.** Confirmed served features line up with chart candles before any strategy was trusted.
3. **Established the baseline.** Long after every green candle, short after every red one, hold exactly one minute. 2,991,516 trades: what the candle alone is worth.
4. **Measured the whale.** Same one-minute bet, taken only when the minute's biggest print reached 100 contracts, in the print's direction. 6,780 trades.
5. **Swept the horizon.** Holding period 1 to 30 minutes, mapping how long the print's information keeps paying before it decays.
6. **Swept the threshold.** Minimum print size 10 to 300 contracts at three horizons: where size starts to mean something, and where it stops.
7. **Split the sessions.** Cash hours against the overnight, same rules.
8. **Located the print.** Wick against body: does it matter where inside the candle the whale executed?
9. **Read the room.** Classified every whale by the resting-book imbalance it arrived into: aligned, opposed, or balanced.
10. **Assembled and stressed the final configuration.** Full trade distribution, per-trade adverse excursion, 1,000 shuffled equity paths, significance and an analytic cost overlay.



Equity Curve Summary

Overall	
Net Profit	1.213.935,00 $
Total # of Trades	2.991.516
Average Trade	0,41 $
Max Close To Close Drawdown	-265.635,00 $
Profit Factor	1,01

FIG. 2 · ANATOMY OF ONE BET: SIGNAL PRINT (TRIANGLE), ENTRY AT NEXT OPEN, TIME EXIT AT THE HORIZON

> *Every result in this paper comes from the same signal file with a different TestMode input. One compiled source reproduces the entire study.*
>
> METHOD NOTE

03

// WHAT THE CANDLE IS WORTH

THE BASELINE.

AVG TRADE	TRADES	PROFIT FACTOR	NET PROFIT
$0.41	**2,991,516**	**1.01**	**$1.21M**
1-MIN HOLD, GROSS	ONE PER SIGNAL CANDLE	ALL TRADES	GROSS, 9.1 YEARS

The control experiment: buy after every green one-minute candle, sell short after every red one, hold exactly one candle. Across almost three million bets there is a real, measurable momentum effect at the one-minute scale, worth **41 cents per contract per trade**.

Two properties matter. First, the magnitude: 41 cents is roughly one fiftieth of a tick's dollar value beyond breakeven, an edge that exists and is completely unharvestable, since round-trip friction is about 35 times larger. Second, the shape: the effect was absent through the low-volatility 2017 to 2019 era, where the strategy bled for roughly the first 700,000 trades, and only emerged afterward. Candle momentum is regime-dependent even before costs.

This number is the yardstick for everything that follows. Any order-flow feature claiming to carry information must beat it under the identical bet: same instrument, same horizon, same fill convention. The comparison is the study.

> *Following the candle is not zero. It is 41 cents: statistically real, economically nothing. The question is what the tape adds on top.*
>
> SECTION CONCLUSION

04

// THE FIRST RESULT

THE BIG TRADE.

AVG TRADE	VS BASELINE	TRADES	PROFIT FACTOR
$1.88	**4.6x**	**6,780**	**1.03**
WHALE ≥100, 1-MIN HOLD	IDENTICAL BET LENGTH	~3 SIGNALS / DAY	GROSS

The core test. Take the identical one-minute bet, but only on minutes whose largest print reached **100 contracts**, and take it in the direction of that print: long after a large aggressive buy, short after a large aggressive sell.

Per bet of identical length, the whale carries **4.6 times the information of the candle**. And the comparison is conservative: the whale's first minute turns out to be its weakest, as the next section shows.

Note what the hit rate does: nothing. Percent profitable sits between 48.7% and 50.9% across every configuration in this study. The edge is not "being right more often". It is winners running slightly further than losers over the same fixed horizon: payoff asymmetry, the signature of genuine directional information rather than of a lucky win-rate pocket.

// READ ON THE MECHANISM

A 100-lot market order is not retail. Prints of that size are overwhelmingly institutional executions, and the aggressor flag on MBO data tells us which side initiated. The hypothesis under test is simply that informed size leaves a wake: that the market does not instantly and fully price what the print reveals. The 4.6 multiple says it does not.

> *Same market, same decade, same one-minute bet. The candle: 41 cents. The whale: a dollar eighty-eight.*

SECTION CONCLUSION

05

// HOW LONG THE INFORMATION LASTS

HORIZON.

THE EDGE NEEDS TIME TO BREATHE.

NQ FUTURES · AVG GROSS PROFIT PER TRADE ($) BY HOLDING PERIOD · PRINTS ≥ 100 LOTS · MAY 2017–JUL 2026 · LEVEL-3 DATA

ORANGE = CLEARS ESTIMATED ROUND-TRIP COSTS (COMMISSION + 1-TICK SLIPPAGE PER SIDE). GROSS BACKTEST, ALWAYS FLAT WHEN NO SIGNAL.

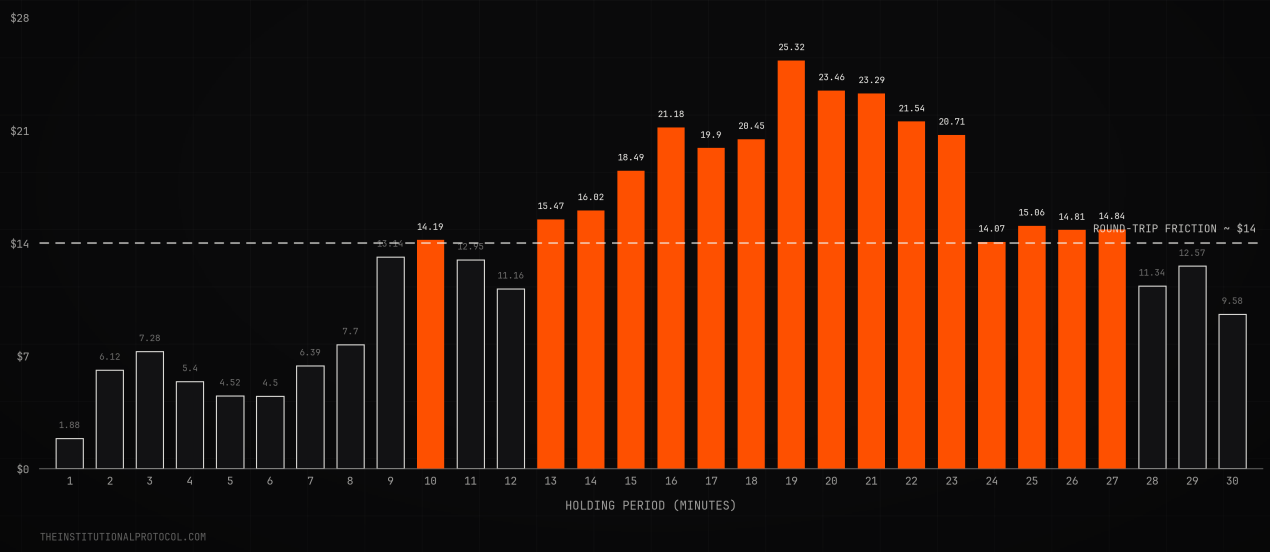


FIG. 3 · AVG GROSS PROFIT PER TRADE BY HOLDING PERIOD, PRINTS ≥100 LOTS. ORANGE BARS CLEAR THE ~$14 FRICTION LINE

Holding the whale bet from 1 to 30 minutes maps the life of the information. The edge builds slowly through the first eight minutes, steps up near minute nine, and forms a **robust plateau from 13 to 23 minutes at $15.47 to $25.32 per trade**, peaking at $25.32 at minute 19. Past minute 23 it decays, reaching $9.58 by minute 30: the market has finished digesting the print.

Two disciplined readings. The tradable claim is the **plateau, not the peak**: ten consecutive holding periods clear estimated round-trip costs, which is the difference between structure and a lucky parameter. And the market takes a quarter of an hour to absorb a block trade, which contradicts the scalper's assumption that order-flow information lives and dies in seconds.

> *The edge needs time to breathe. It is smaller than the spread at one minute, and larger than the spread for every holding period from 13 to 23.*
>
> SECTION CONCLUSION

06

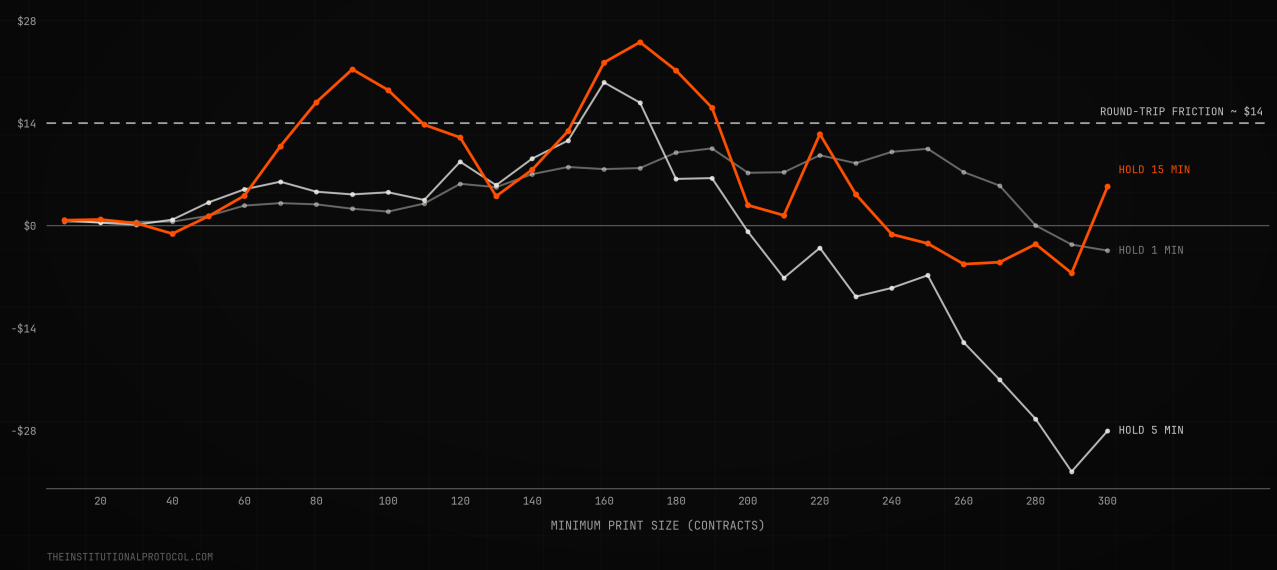
// WHICH SIZE CARRIES THE EDGE

THE THRESHOLD STUDY.

WHERE SIZE BECOMES INFORMATION.

NQ FUTURES · AVG GROSS PROFIT PER TRADE ($) BY MINIMUM PRINT SIZE · MAY 2017–JUL 2026 · LEVEL-3 DATA

THREE HOLDING PERIODS. EDGE APPEARS NEAR 60 LOTS, PEAKS AT 70-190, AND DIES ABOVE ~200. GROSS BACKTEST.



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FIG. 4 · AVG GROSS PROFIT PER TRADE BY MINIMUM PRINT SIZE, THREE HOLDING PERIODS, SWEEP 10 TO 300 CONTRACTS

Information onset is near 50 to 60 lots. Below that, nothing: thresholds of 10 to 40 contracts average $0.28 to $0.80 at every horizon, statistically indistinguishable from the candle baseline. A "biggest print" of 20 lots is just the bar talking.

The working band is roughly 70 to 190 contracts. Through this zone both the per-trade edge and the hit rate rise with size: at 15-minute holds, $10.82 at 70 lots, $21.34 at 90, peaking at $25.05 at 170, while percent profitable climbs from 45% to almost 53%. A clean dose-response relationship: more size, more information.

Above roughly 200 contracts the edge dies, and at some horizons inverts: at 5-minute holds, thresholds of 210 to 300 average between minus $7 and minus $34 per trade. Sample sizes out there are small (under 1,300 trades), so the inversion is suggestive rather than proven, but the loss of positive edge is consistent across all three horizons. The reading: the mega-print is a different animal. Negotiated blocks, stop cascades and climactic prints reprice the market instantly, then partially revert. **Size is information up to a point. Past it, size is exhaustion.**

A note on discipline: this study's headline threshold of 100 contracts sits inside the band but below its optimum at every horizon. It was fixed before this sweep was run, and it stays fixed. Selecting the sweep's peak after seeing it would be curve-fitting, which is the failure mode this document exists to model against.

07

// CASH HOURS VS THE OVERNIGHT

SESSIONS.

RTH SHARE OF PRINTS	RTH AVG, 1-MIN	RTH AVG, 15-MIN	OVERNIGHT AVG, 1-MIN
79%	**$4.16**	**$24.31**	**−$6.65**
5,348 OF 6,780	VS $1.88 FULL SESSION	PF 1.11	DERIVED BY SUBTRACTION

Restricting entries to 09:30 to 16:00 Eastern concentrates the signal twice over. First, mechanically: 79% of all 100-lot prints occur inside the 6.5 cash hours. Second, qualitatively: the per-trade edge more than doubles at the one-minute horizon, and the one-minute RTH equity curve carries a drawdown of only $7,215 against $22,250 of profit.

The overnight residual, derived by subtracting the RTH run from the full-session run, is slightly negative at both horizons: about minus $6.65 per trade at one minute and minus $2.20 at fifteen.

The interpretation is one of the cleanest in the study. Overnight, a 100-lot print is proportionally enormous against the thin tape, yet it predicts nothing. That is consistent with mechanical flow: rolls, spread legs, forced hedging. Informed size trades when it can hide inside volume, and that is the cash session. **The whale edge is a cash-hours phenomenon, entirely.**

> *Overnight, the biggest print in the minute is usually plumbing. In cash hours, it is usually a decision.*
>
> SECTION CONCLUSION

08

// WHERE INSIDE THE CANDLE

WICK OR BODY.

BODY PRINT, 1-MIN	BODY PRINT, 15-MIN	WICK PRINT, 1-MIN	WICK PRINT, 15-MIN
−$0.14	**$8.05**	**$4.31**	**$34.13**
3,705 TRADES	BELOW FRICTION	3,075 TRADES	PF 1.16

Every whale print falls either inside the candle's body or inside one of its wicks. The two subsets partition the 6,780 whale trades exactly, and they could not behave more differently.

The **body print is dead**: minus 14 cents at one minute, a statistical zero across 3,705 trades, and only $8.05 even at fifteen minutes. The **wick print carries everything**: $4.31 at one minute and $34.13 at fifteen, with a profit factor of 1.16, the strongest fifteen-minute figure in this study.

A popular retail hypothesis holds the opposite: that a print inside the wick is "rejection" to be faded, while a print inside the body is "acceptance" confirming direction. The data inverts it. Follow the wick whale; ignore the body whale.

// MECHANISM

A large buy printed in the lower wick is size absorbing a flush at the extreme: the whale was filled at the low and the bar closed away from it. That is the signature of informed execution at favorable prices, sweeping liquidity where it pooled. The body print, by contrast, is size paying market impact mid-range, chasing. Where the whale chose to execute tells you what kind of whale it was.

> *Location beats size refinement: at the same 100-lot threshold, the wick subset quadruples the pooled fifteen-minute edge of the body subset.*

SECTION CONCLUSION

09

// THE CONTEXT THE PRINT ARRIVES INTO

THE RESTING BOOK.

Level 3 data lets us do one more thing no candle can: read the resting order book the whale traded into. For every whale minute we compute the depth imbalance across the top 50 price levels per side, and classify the book as **aligned** (leaning the print's way, imbalance beyond 10%), **opposed** (leaning against it), or **balanced** (inside 10%).

WHALE INTO ...	AVG / TRADE (1-MIN)	TRADES
Aligned book	-$4.34	783
Opposing book	-$11.21	1,170
Balanced book	+$6.47	4,557

The whale predicts **only when the book is balanced**. Into a leaning book, in either direction, the print's predictive power disappears; the opposed case, size slamming into a wall of resting orders that absorbs it, reads like classic failed aggression. The balanced-book subset is the study's best-behaved signal: profit factor 1.11 at one minute with a $7,440 drawdown, and a holding-period sweep that clears the $14 friction line **continuously from minute 9 through minute 30**, peaking at $34.97 at minute 19, with no decay back below costs inside the tested range.

	HoldBars (OFB_BigTradeStudy_SIG)	Avg Trade	Net Profit	Gross Profit	Gross Loss	Total Trades	% Profitable
1	1	6,47	29.500,00	302.915,00	-273.415,00	4.557	49,31 %
2	2	12,40	54.700,00	417.305,00	-362.605,00	4.413	49,63 %
3	3	14,06	60.715,00	493.995,00	-433.280,00	4.318	49,44 %
4	4	11,48	48.945,00	547.530,00	-498.585,00	4.262	49,01 %
5	5	11,30	47.485,00	585.225,00	-537.740,00	4.201	48,39 %
6	6	10,33	42.900,00	623.925,00	-581.025,00	4.154	49,28 %
7	7	12,44	51.220,00	668.670,00	-617.450,00	4.119	49,48 %
8	8	13,87	56.565,00	710.215,00	-653.650,00	4.079	49,03 %
9	9	17,53	70.920,00	740.730,00	-669.810,00	4.046	49,53 %
10	10	23,04	92.490,00	775.775,00	-683.285,00	4.015	50,98 %
11	11	19,71	78.715,00	787.645,00	-708.930,00	3.993	50,66 %
12	12	21,21	84.090,00	819.150,00	-735.060,00	3.965	50,82 %
13	13	23,04	90.870,00	849.325,00	-758.455,00	3.944	50,84 %
14	14	22,78	89.265,00	870.665,00	-781.400,00	3.919	50,45 %
15	15	25,38	98.820,00	892.660,00	-793.840,00	3.894	50,74 %
16	16	28,59	110.480,00	919.745,00	-809.265,00	3.864	51,11 %
17	17	26,10	100.365,00	923.705,00	-823.340,00	3.845	51,29 %
18	18	28,92	110.600,00	946.130,00	-835.530,00	3.825	50,93 %
19	19	34,97	133.135,00	964.340,00	-831.205,00	3.807	51,14 %
20	20	29,70	112.590,00	972.380,00	-859.790,00	3.791	50,65 %
21	21	30,41	114.670,00	993.445,00	-878.775,00	3.771	50,68 %
22	22	30,26	113.885,00	1.012.230,00	-898.345,00	3.763	50,36 %
23	23	28,52	106.880,00	1.017.590,00	-910.710,00	3.747	50,07 %
24	24	21,88	81.640,00	1.003.295,00	-921.655,00	3.731	50,12 %
25	25	24,93	92.640,00	1.028.425,00	-935.785,00	3.716	49,76 %
26	26	23,49	87.075,00	1.037.605,00	-950.530,00	3.707	49,96 %
27	27	23,71	87.525,00	1.050.885,00	-963.360,00	3.692	49,92 %
28	28	19,35	71.215,00	1.056.540,00	-985.325,00	3.680	50,63 %
29	29	20,46	75.110,00	1.079.080,00	-1.003.970,00	3.671	50,45 %
30	30	19,54	71.370,00	1.091.725,00	-1.020.355,00	3.653	49,79 %

FIG. 5 · BALANCED-BOOK WHALE, HOLDING PERIOD SWEEP 1 TO 30 MINUTES. THE EDGE CLEARS COSTS FROM MINUTE 9 ONWARD

// DISCLOSURE

The wick filter and the balanced-book filter were both discovered inside this sample, through this test sequence. They are reported as findings, and the final configuration that uses one of them should be read with that in mind. They likely describe the same phenomenon from two angles: the informative whale is the one executing at an extreme, into a book that had not tipped its hand. Stacking both filters is deliberately left as future work rather than mined here.

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// EVERYTHING THE STUDY LEARNED, IN ONE SIGNAL

THE FINAL CONFIGURATION.

// SPECIFICATION

Signal	Whale into balanced book
Print threshold	≥ 100 contracts
Book filter	|imbalance| < 10%, top 50 levels
Session	RTH, 09:30–16:00 ET
Hold	15 minutes, fixed
Direction	With the print
Size	1 contract, no compounding
Stops / targets	None. Time exit only

AVG TRADE
$26.85
GROSS

TRADES
3,171
1,655 L / 1,516 S

PROFIT FACTOR
1.13
NET PROFIT $85,145

MAX DD
$16,530
CLOSE TO CLOSE

About 1.4 signals per trading day. Long and short contribute almost identically ($26.18 long, $27.58 short), which matters: the edge is not borrowed from the index's decade-long drift.

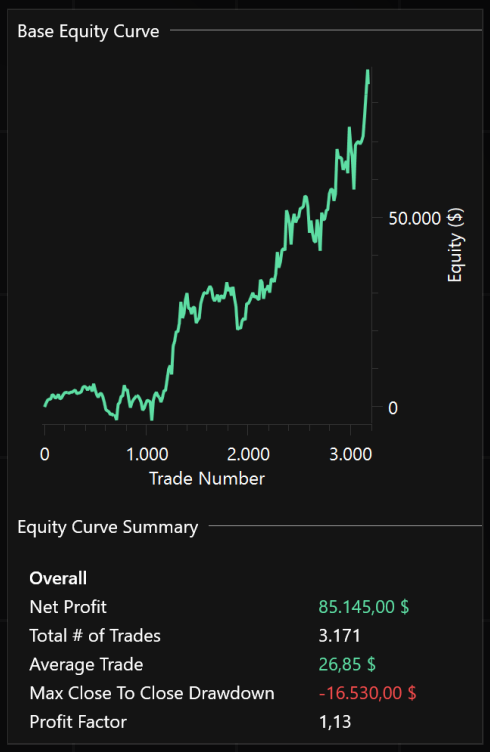


FIG. 6 · EQUITY CURVE AND SUMMARY, 3,171 TRADES



FIG. 7 · DETAILED EQUITY, $100K INITIAL, MAY 2017 TO JUL 2026. THE CURVE EARNS ITS SLOPE ACROSS REGIMES, WITH THE STRONGEST STRETCH IN THE HIGH-VOLATILITY YEARS

11

// PER-TRADE GEOMETRY

DISTRIBUTION & EXCURSION.

Strategy Analysis | Trade Analysis | Periodical Analysis | Monte-Carlo Analysis | Graphs | Settings

Total # of Trades	Number Winning Trades	Number Losing Trades	Avg Trade (win & loss)	Ratio Avg Win / Avg Loss
3.171	1.613	1.538	$26,85	1,07
1655 long · 1516 short	50,87% profitable	49,13% of trades	All trades	All trades

List of Trades | Total Trade Analysis | Outliers | Run-up/Drawdown | Trade Series Analysis | Trade Series Statistics

Metric	All trades	Long trades	Short trades
TRADE SUMMARY			
Total # of Trades	3.171	1.655	1.516
Total # of Open Trades	0	0	0
Number Winning Trades	1.613	844	769
Number Losing Trades	1.538	796	742
Percent Profitable	50,87%	51,00%	50,73%
TRADE PROFITABILITY			
Avg Trade (win & loss)	$26,85	$26,18	$27,58
Average Winning Trade	$468,09	$448,32	$489,79
Average Losing Trade	($435,56)	($420,92)	($451,26)
Ratio Avg Win / Avg Loss	1,07	1,07	1,09
Largest Winning Trade	$14.765,00	$14.765,00	$4.055,00
Largest Losing Trade	($4.285,00)	($4.235,00)	($4.285,00)
TRADE DURATION (BARS)			
Avg # Bars in All Trades	15,40	15,40	15,40
Avg # Bars in Winning Trades	15,50	15,50	15,50
Avg # Bars in Losing Trades	15,30	15,40	15,20
Avg # Bars in Even Trades	15,80	16,00	15,20

TRADE STATS

BEST / WORST

Largest Win	$14.765,00
Largest Loss	($4.285,00)
Avg Win $	$468,09
Avg Loss $	($435,56)
Win/Loss ratio	1,07

RUN-UP

Avg Run-up $	$473,21
Max Run-up $	$16.145,00

DRAWDOWN

Avg DD $	($441,73)
Max DD $	($5.200,00)

OUTLIERS

Number of Outliers	336
Outlier P&L	$11.410,00

DISPERSION

Standard Deviation	$745,20

STREAKS

Max Consec. Winners	11
Max Consec. Losers	8
Largest Consec. Profit	$15.050,00
Largest Consec. Loss	($7.490,00)

FIG. 8 · TOTAL TRADE ANALYSIS. WIN RATE 50.87%, AVG WIN $468 AGAINST AVG LOSS $436, LARGEST WIN $14,765 AGAINST LARGEST LOSS $4,285

The geometry is the healthy kind. A coin-flip hit rate (50.87%) paired with a payoff ratio of 1.07 and a right tail three times longer than the left: the largest winner is $14,765 against a largest loser of $4,285, and the outlier population as a whole contributes positively (+$11,410 across 336 outliers). No single trade carries the P&L; the edge is diffuse, which is what a structural effect should look like.

Adverse excursion. The average trade endures a maximum adverse excursion of about $442 on its way to an average result of $27, and the worst single excursion is $5,200. That is the empirical case for the no-stop design: a stop tight enough to be felt would be hit inside the noise band of a typical winner and would convert edge into churn. Risk here is managed by the horizon and by size, not by a stop placed inside the excursion cloud.

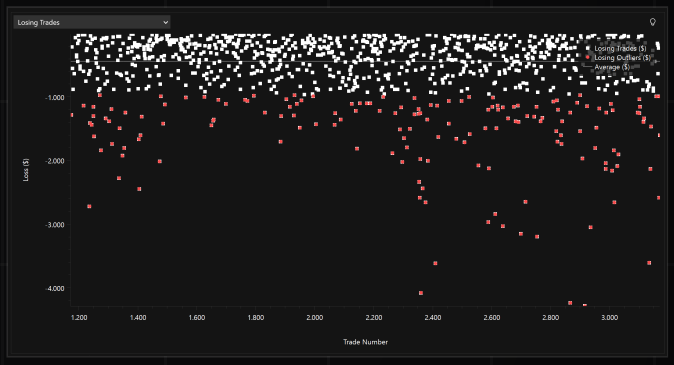


FIG. 9 · THE LOSS TAIL. LOSSES CLUSTER SHALLOW; RED OUTLIERS THIN OUT SMOOTHLY TOWARD THE -$4K WORST CASE, WITH NO CLIFF

12

// STRESS, SIGNIFICANCE, COSTS

SHUFFLES, SIGNIFICANCE, COSTS.

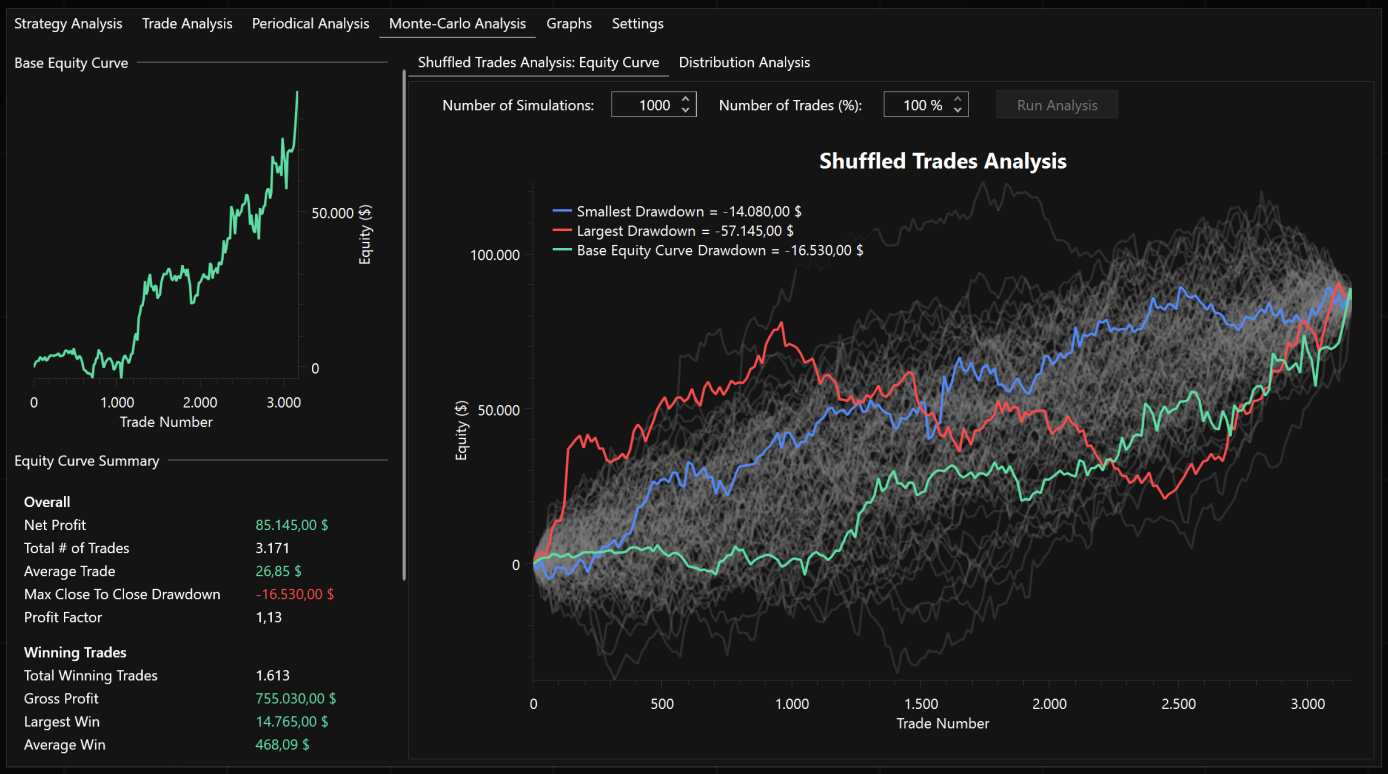


FIG. 10 · 1,000 SHUFFLED TRADE SEQUENCES. REALISED DRAWDOWN $16,530 SITS NEAR THE BOTTOM OF THE SHUFFLED RANGE ($14,080 TO $57,145)

// Path vs set

Shuffling the 3,171 trades a thousand times and re-walking the equity curve asks whether the realised drawdown was luck of the ordering. It partly was: the base curve's $16,530 sits near the friendly end of the shuffled range, whose worst ordering reaches $57,145. A deployer should budget for the tail of that distribution, not for the realised number. Terminal profit is invariant by construction; the path is not.

// Significance

With a per-trade standard deviation of $745.20 across 3,171 trades, the $26.85 mean carries a t-statistic near **2.0**: significant at conventional levels, and honestly reported as no stronger than that. Because the configuration's filters were found in this same sample, the true out-of-sample expectancy is plausibly lower. Walk-forward confirmation is listed as the first next check.

// Cost overlay

Round-trip friction assumption	$14 / trade
Total drag, 3,171 trades	$44,394
Net profit after friction	$40,751
Avg trade after friction	$12.85
Profit factor after friction	~1.06

Friction modelled as commission plus one tick of slippage per side, applied analytically to every trade. The configuration survives costs with roughly half its gross edge intact. It survives because of the fifteen-minute horizon: the identical signal harvested at one minute would hand the entire edge to the spread.

> *The information is real at every horizon. The edge, net of the spread, exists only where the study said it would: where the signal is given time.*

SECTION CONCLUSION

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// WHAT THE EVIDENCE SUPPORTS

VERDICT.

INFORMATION CONFIRMED | EDGE SURVIVES COSTS AT 15 MIN | T ≈ 2.0

// What the evidence shows

Large trades predict. 4.6 times the candle's information per identical bet, built on payoff asymmetry rather than hit rate, expressed over 15 to 20 minutes.

The edge has structure. It lives in prints of 70 to 190 contracts, in cash hours, in wick executions, in balanced books. Each slice was measured, not assumed, and the slices agree with one economic story: the informative whale is the one the market did not see coming.

The final configuration clears costs across a wide holding-period band, with a drawdown profile that survives 1,000 reorderings of its own trades.

// Next checks before capital

Walk-forward on rolling windows; split-half stability of the balanced-book filter; slippage stress at two ticks per side; regime decomposition by volatility bucket; and live paper confirmation that fills at the next-bar open are achievable at signal times.

// Statement of limitations

Not financial advice. This document is a statistical study prepared for the members of The Institutional Protocol and does not constitute an offer, solicitation, or recommendation to trade any instrument. All headline results are gross of costs except where a cost overlay is explicitly stated; the overlay is analytic, not a re-execution. The wick and balanced-book filters were identified within the studied sample; out-of-sample performance may be materially lower. Fixed-horizon results assume market-order fills at the next one-minute open on a per-contract basis; capacity, latency and queue effects at real size are untested. Shuffled-sequence analysis assumes exchangeable trades; volatility clustering may understate tail drawdowns. The sample begins where CME MBO history begins (May 2017) and covers a single instrument. Every figure is reproduced from MultiCharts or from the study's own render pipeline.

> *One dataset, one measurement, nine tests. The candle is the shadow. The tape is the object casting it.*
>
> CLOSING COMMENTARY

// APPENDIX A · FULL SOURCE, PAGE 1 OF 2

THE CODE.

The complete PowerLanguage signal behind every table and chart in this paper. Copy it into your MultiCharts editor and run it against The Bridge, shared inside the extra classes of The Institutional Protocol. Each TestMode reproduces one section of this study.

```
// ============================================================================
// OFB_BigTradeStudy_SIG  -  master signal behind every test in this paper.
// Fixed-length exits: every trade is exactly HoldBars bars, re-entry allowed.
//
//   TestMode 1 = candle only (baseline)
//   TestMode 2 = whale only (biggest print ≥ MinSize)
//   TestMode 3 = book imbalance only (|imb| ≥ Threshold)
//   TestMode 4 = whale AGAINST candle
//   TestMode 5 = whale into ALIGNED book     (print dir = book lean)
//   TestMode 6 = whale into OPPOSING book    (print dir <> book lean)
//   TestMode 7 = whale into FLAT book        (|imb| < Threshold)
//   TestMode 8 = whale printed in candle BODY
//   TestMode 9 = whale printed in candle WICK
//
// Requires the OFB function library compiled (ships with The Bridge).
// ============================================================================
inputs:
    TestMode( 7 ),
    MinSize( 100 ),        // whale threshold, contracts
    Levels( 50 ),          // book depth per side: 5 | 10 | 20 | 50
    Threshold( 10 ),       // book imbalance % trigger
    HoldBars( 15 ),
    UseRTH( true ),        // true = entries only StartTime-EndTime
    StartTime( 930 ),
    EndTime( 1600 ),
    Qty( 1 );

variables:
    Blk( 0 ), Sz( 0 ), PrintDir( 0 ), Px( 0 ),
    BodyHi( 0 ), BodyLo( 0 ), InBody( false ),
    BidSz( 0 ), AskSz( 0 ), Imb( 0 ), LiqDir( 0 ),
    ColorDir( 0 ), Dir( 0 );

// ---- features ---------------------------------------------------------
Blk = OFB_BigTrade;
Sz  = AbsValue( Blk );
PrintDir = 0;
if Sz ≥ MinSize then begin
    if Blk > 0 then PrintDir = 1 else PrintDir = -1;
end;

Px = OFB_BigTradePx;
BodyHi = MaxList( Open, Close );
BodyLo = MinList( Open, Close );
InBody = Px ≥ BodyLo and Px ≤ BodyHi;

BidSz = OFB_BidDepth( Levels );
AskSz = OFB_AskDepth( Levels );
Imb = 0;
if BidSz + AskSz > 0 then
    Imb = ( BidSz - AskSz ) / ( BidSz + AskSz ) * 100;
LiqDir = 0;
```

// APPENDIX B · FULL SOURCE, PAGE 2 OF 2

```
if Imb ≥  Threshold then LiqDir =  1;
if Imb ≤ -Threshold then LiqDir = -1;

ColorDir = 0;
if Close > Open then ColorDir =  1;
if Close < Open then ColorDir = -1;

// ---- desired position, reset every bar -----------------------------------
Dir = 0;
if TestMode = 1 and ColorDir <> 0 then Dir = ColorDir;
if TestMode = 2 and PrintDir <> 0 then Dir = PrintDir;
if TestMode = 3 and LiqDir <> 0 then Dir = LiqDir;
if TestMode = 4 and PrintDir <> 0 and ColorDir <> 0
   and PrintDir <> ColorDir then Dir = PrintDir;
if TestMode = 5 and PrintDir <> 0 and PrintDir = LiqDir then Dir = PrintDir;
if TestMode = 6 and PrintDir <> 0 and LiqDir <> 0
   and PrintDir <> LiqDir then Dir = PrintDir;
if TestMode = 7 and PrintDir <> 0 and LiqDir = 0 then Dir = PrintDir;
if TestMode = 8 and PrintDir <> 0 and Px > 0 and InBody then Dir = PrintDir;
if TestMode = 9 and PrintDir <> 0 and Px > 0 and InBody = false then
    Dir = PrintDir;

if UseRTH and ( Time < StartTime or Time ≥ EndTime ) then
    Dir = 0;

// ---- orders --------------------------------------------------------------
if Dir = 1 then
    Buy ( "StudyLE" ) Qty contracts next bar at market;
if Dir = -1 then
    SellShort ( "StudySE" ) Qty contracts next bar at market;

// ---- fixed-length time exit ----------------------------------------------
if MarketPosition = 1 and BarsSinceEntry ≥ HoldBars - 1 then
    Sell ( "TimeX.L" ) next bar at market;
if MarketPosition = -1 and BarsSinceEntry ≥ HoldBars - 1 then
    BuyToCover ( "TimeX.S" ) next bar at market;
```

// HOW TO RUN IT

Requires MultiCharts 64-bit and The Bridge with the OFB function library compiled (both provided, with setup instructions, inside the extra classes of The Institutional Protocol). Chart the Bridge-exported NQ series at 1 minute, apply the signal, set commission and slippage to zero for the gross tests, and select the TestMode for the section you want to reproduce. Optimize HoldBars or MinSize to reproduce the sweeps. Access to The Bridge is available exclusively through an Institutional Protocol license, as a member of The Institutional Protocol.

// A NOTE TO MEMBERS

THE BRIDGE.

Every test in this paper ran on official CME Level 3 data: the order-by-order record that, until today, only institutional desks could realistically buy, store, rebuild and research. The Bridge changes that. It reconstructs the full order book, distills it into features you can read in plain PowerLanguage, and serves the same engine to your backtests and your live charts, so what you research is what you run. Years of testing on complex microstructure data, with no coding experience required. For retail, this simply was not possible until now.

Congratulations to all the members of The Institutional Protocol, who have access to the complete set of order-flow features derived from MBO / Level 3 data: large trades, tape-exact VWAP and value areas, resting-book depth and imbalance, iceberg detection and more, ready to be used to develop, test and automate your own trading strategies.

The Bridge is available exclusively with an Institutional Protocol license, inside the extra classes of the program.

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